

MedAssist

Team Name:	Orbit
Team Leader:	Joylan Dsouza
Team Members:	Manvith, Amaan, Lakshya, Denzil, Clinston
College/University:	St Joseph Engineering College, Mangalore
Selected Track:	Agentic AI For Billions

1. Problem Statement

Patients discharged from hospitals or handed a prescription routinely face three disconnected problems: they don't understand what their medicines are for or how to take them; they can't easily tell if a cheaper generic alternative exists; and they don't know whether their local pharmacy actually has the medicine in stock before they travel there.

This especially affects people managing multiple medications, elderly patients, and families in Tier 2/3 cities where pharmacy stock information isn't digitized. Poor medicine adherence and unnecessary pharmacy trips are common, everyday failures that affect nearly everyone who has ever left a hospital with a discharge summary.

2. Proposed Solution

MedAssist is an AI-powered portal that turns a prescription or discharge summary into clear, actionable guidance across four connected features, all built around a single AI engine that reads unstructured medical text and extracts structured information. The agent does not diagnose — it interprets what a doctor has already prescribed, keeping the system safe and explainable.

- **Explain My Discharge** — converts medical text into a plain-language, day-by-day medicine schedule with follow-up questions for the doctor
- **Find Cheaper Medicine** — identifies generic equivalents and estimated cost savings
- **Check Before You Go** — shows nearby pharmacy stock status and best pickup time, flagging known drug interactions
- **Scan a Medicine** — barcode scanning to check expiry status and whether a medicine treats a common or rare condition

3. Target Users & Impact

Primary users are patients and families managing prescriptions after a hospital visit or doctor consultation — particularly the elderly, people on multiple medications, and families in Tier 2/3 towns with limited digital pharmacy infrastructure. By making medicine information clear, affordable alternatives visible, and pharmacy trips efficient, MedAssist reduces medication errors, saves money for financially vulnerable families, and cuts wasted time, serving a need common to nearly every patient in India.

4. Technology Stack

Frontend: HTML, CSS, and JavaScript as a single-page portal with a shared interface across all four features. AI: an LLM API powers the extraction engine that turns prescription or discharge text into structured data (medicine name, dosage, frequency, duration). Data: a demo dataset of medicines and

pharmacies (prices, stock, generic mappings) supports the substitution and pharmacy-check features. Hardware interface: a browser-based barcode scanning library for the scan feature.

5. Innovation & Scalability

MedAssist's innovation is combining four everyday pharmacy/prescription pain points into one AI engine rather than four separate tools, and showing its reasoning at each step so results are transparent rather than a black box. It stays deliberately outside clinical diagnosis, which keeps it safe to deploy widely. The system scales by connecting the demo dataset to real pharmacy inventory APIs and hospital discharge systems, and can extend to more regional languages and a mobile app for wider, low-bandwidth access.

6. Prototype / References

Prototype link / wireframe / GitHub repository: *to be added*

7. Team Contribution

Member	Role & Contribution
Joylan Dsouza	Team lead; AI prompt engineering and the core text-extraction engine; overall coordination
Manvith	Frontend development — portal UI, navigation, and styling across all four features
Amaan	Find Cheaper Medicine and Check Before You Go feature logic; demo dataset design
Lakshya	Explain My Discharge feature logic and medicine schedule generation
Denzil	Barcode scanning integration and the Scan a Medicine feature
Clinston	Testing, dataset curation, documentation, and demo preparation